Adele Chinda

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Professional Summary

Ph.D. candidate in Computer Science at **Georgia State University** working on **LLM agents**, **multimodal foundation models**, and **diffusion-based generation**, with **two first-author publications** and an industry research internship evaluating and red-teaming production LLM systems. I work on making agents **reason reliably over long horizons**: rigorous empirical evaluation, controlled ablations, and efficient, controllable generation — from hypothesis to working system.

Research and Professional Experience

AI Research Intern

Jun 2026 – Aug 2026

Lenovo | Qira — Chicago, IL

- Conducted a systematic empirical evaluation of **13 LLMs** across the Gemini, Llama, Mistral, and Qwen families (**70+ scored runs** over a 10-prompt, 5-tier suite with ground-truth annotations), building an automated auto-scoring harness that quantified plan quality and latency and yielded a reproducible model leaderboard (top free model: Llama 4 Scout, ∼2.2s latency, 95.2 quality).
- Designed a provider-agnostic LLM evaluation framework, a unified abstraction layer (base class, factory, adapters) over OpenAI-compatible, Anthropic, Groq, and local Ollama backends, enabling new models to be benchmarked from a single config entry, removing engineering overhead from controlled cross-model experiments.
- Built an end-to-end **agentic** pipeline (task planning + browser execution) and hardened it for reliable multi-step task completion via strict JSON-schema validation, tool-call argument coercion, and rate-limit retry logic, studying failure modes in long-horizon tool use.
- Ran a controlled context-optimization study on live multi-step web tasks, capturing per-step token telemetry that quantified **28–43% token savings** and produced a data-driven technical proposal recommending an optimal sliding-window size.
- Led a **red-teaming / failure analysis** of context grounding and meeting-summarization: constructed an adversarial knowledge base (conflicting versions, temporal updates, intentional evidence gaps) and an ASR-error stress test, surfacing **10 documented defects** with reproduction steps and proposed fixes, including a *hallucination-under-missing-evidence* failure central to model reliability and alignment.
- Translated findings into roadmap impact, presenting calibration and routing failures (e.g., trivial queries mis-routed to costly browser planning) to product and engineering stakeholders.

AI Ph.D. Researcher

Sep 2025 – Present

Advanced Mobility & Augmented Intelligence Lab, Georgia State University | Advisor: Dr. Haoxin Wang — Atlanta, GA

- Developed **DIAL**, a training framework for block-size-elastic diffusion language models (DLMs) using mixed block-size training and per-block noise conditioning, enabling dynamic inference-time quality–compute trade-offs from a single parameter set without retraining.
- Created **ATAT**, an adaptive masking scheme for discrete DLMs using per-token time-warping based on self-supervised surprisal scores, achieving a **7.4% perplexity reduction** over uniform corruption baselines.
- Designed **ViMoE**, a sparse mixture-of-experts vision–language model with dynamic token-level routing that matched dense-model accuracy at ∼**3**× **fewer active parameters** (first-author publication).
- Built **SEEK-Multi**, a multi-agent LLM framework where specialized agents collaboratively reason over complex queries, **outperforming single-agent baselines by 18%** on multi-hop reasoning (first-author publication).
- Studying **scaling behavior** of multimodal MoE architectures — how expert count, routing, and modality-specific capacity affect quality — training on multi-GPU clusters with PyTorch FSDP, DeepSpeed ZeRO, and bf16 mixed precision.

AI Ph.D. Researcher

Aug 2024 – Sep 2025

Assistive Intelligence Lab, Georgia State University | Advisor: Dr. Yi Ding — Atlanta, GA

- Developed LLM-augmented planning architectures for embodied agents, integrating chain-of-thought reasoning with goal-conditioned action selection in partially observable environments.
- Built spatial-semantic memory modules that **improved multi-step navigation task completion by 12%** in AI2-THOR.
- Engineered data pipelines converting AI2-THOR and Habitat simulation trajectories into supervised fine-tuning datasets for LLM-based planners, accelerating training and experimentation.

Undergraduate Research Assistant — *Sep 2023 – Jan 2024*

Artificial Intelligence Lab, Belgorod State University — Belgorod, Russia

- Developed intelligent control algorithms for non-linear robotic systems, improving precision and system stability.
- Collaborated with supervisors on data-driven modeling for adaptive robotic performance in uncertain conditions.

ML Research Intern — *Jun 2023 – Aug 2023*

CVisionLab — Rostov, Russia

- Developed probabilistic **uncertainty-estimation** methods for visual perception models, benchmarking Bayesian against fuzzy-logic approaches and showing improved classification robustness under distribution shift.
- Built evaluation pipelines comparing **model calibration** and out-of-distribution detection across architectures, identifying the most reliable configuration.

Education

2024–2029 **Ph.D. in Computer Science**, Georgia State University
Advisors: Dr. Haoxin Wang (current), Dr. Yi Ding

2020–2024 **B.Sc. in Applied Computer Science**, Belgorod State University, Russia — GPA: 3.89/4.0 (*summa cum laude*)
Supervisor: Assoc. Prof. Evgenia Bolgova

Selected Publications

- **ViMoE: Vision Mixture of Experts with Multimodal Context Awareness**
  A. Chinda (first author). *World Journal of Advanced Research and Reviews*, 2025. Peer-reviewed. [Project Page] [Code]
- **SEEK-Multi: Collaborative Multi-Agent Semantic Reasoning**
  A. Chinda (first author). *International Journal of Science and Research Archive*, 2025. Peer-reviewed. [Project Page] [Code]
- **Contrastive Deep Learning for Variant Detection in Wastewater Genomic Sequencing**
  A. Chinda*, R. Azumah*, H. D. Venkateswara (*equal contribution). Preprint, 2025. [arXiv]

Selected Research Projects

HierarchicalVLM: Efficient Long-Context Video Understanding via Hierarchical Temporal Aggregation

Project Page — Developed a hierarchical temporal aggregation framework for efficient long-context video understanding using vision–language models.
A. Chinda, D. Emeka, M. Koya, N. N. Ezekwem.

GraPO: Graph-based Prediction of Power Outages
Project Page — Developed a machine learning framework leveraging spatiotemporal graph neural networks to

predict power outages during extreme weather events.
A. Chinda, H. Chai, J. Morris, J. Fassett.

Open Source Contributions

- **harmonic** — semantic conflict-resolution framework for multimodal guided diffusion, exploring guidance-time intervention in diffusion sampling. [Code]
- **temporal-fuse** — adaptive long-video understanding with efficient attention and domain-specific adaptation. [Code]
- **adafuse-tts** — adaptive noise scheduling for audio synthesis. [Code]

Technical Skills

- **Multimodal AI & Vision–Language:** ViT, CLIP, Mixture of Experts (MoE), cross-modal attention, visual grounding, video understanding.
- **Diffusion Models:** DDPM, latent diffusion, score-based generative models, classifier-free guidance, diffusion for text generation.
- **Deep Learning & NLP:** Transformers, LLMs (GPT, LLaMA, Mistral, Qwen, Gemini), RLHF, DPO, LoRA/QLoRA fine-tuning.
- **LLM Evaluation & Alignment:** Benchmark design, auto-scoring harnesses, red-teaming, hallucination & calibration analysis, reward modeling.
- **Reinforcement Learning:** MDPs, PPO, SAC, DQN, imitation learning, reward modeling.
- **Multi-Agent & Agentic Systems:** Collaborative reasoning, tool-augmented LLM agents, long-horizon task planning, browser automation.
- **Frameworks & Libraries:** PyTorch, TensorFlow, Hugging Face Transformers & Diffusers, scikit-learn, DeepSpeed, PEFT, Weights & Biases.
- **Simulation & Embodied AI:** AI2-THOR, Habitat, CARLA, Gazebo.
- **Infrastructure & Tools:** Git, Docker, Linux, SLURM, CUDA, distributed training (DDP, FSDP), LaTeX.
- **Programming:** Python, C++, SQL, Bash.

Leadership & Service

Lead, ACM Student Chapter, Georgia State University — organize biweekly technical talks and events.

Member: AMAI Lab, Collaborative Human-AI Center (CHAI), ML Collective, Black in AI, NSBE.